

MARHA MIDHATIEY

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Professional Summary

AI/ML Engineer and System Analyst with a Master's in AI and strong technical expertise across the full SDLC, database architecture, and LLMs. Experienced in leading client requirement gathering (URS), UAT/FAT execution, legacy system upgrades, fine-tuning multi-billion parameter models on HPC clusters and Prompt Engineering.

Technical Skills

Languages & Frameworks: Python, PyTorch, C++, C#, Java, PHP, Visual Basic, .NET, LaTeX, SQL, Hugging Face, TensorFlow, Postman, REST APIs

Functional & SDLC: Requirement Gathering (URS), As-Is/To-Be Workflow Mapping, UML (ERD/DFD), UI/UX Wireframing, FAT/UAT Testing, Defect Tracking (Jira), User Manuals & Documentation

AI / Machine Learning: LLM Fine-Tuning (QLoRA, SFT, LoRA), Neurosymbolic AI, Computer Vision (CNNs, Transfer Learning), MelBandRoformer, Prompt Engineering, Deep Learning, NLP, Transformers, RAG, Vector Databases (FAISS)

Databases & Tools: Oracle DB, MySQL, Power BI, Kafka, Gradio, RapidMiner, Git, Linux HPC Clusters, Docker, Kubernetes, GCP, Azure, Microsoft Office & 365 (Excel, PowerPoint, Word), Figma, Drawio

Technical Experience

Research Assistant (Data Science Lab) | GIST, South Korea (2024 – 2026)

- Researched hybrid AI architectures combining neurosymbolic methods and LLMs for complex visual reasoning.
- Optimized fine-tuning of 7B/13B/32B LLMs on HPC clusters via **QLoRA, SFT, and LoRA**, significantly reducing memory footprint and accelerating throughput.
- Authored research papers and formatted complex mathematical models and algorithms in **LaTeX** for journal and conference submissions.

System Analyst & Database Engineer (Intern) | 3F Resources Sdn. Bhd., Malaysia (2023)

- Elicited functional requirements and authored URS through direct client alignment meetings with clients/ stakeholders.
- Modeled As-Is/To-Be workflows and UML database structures (ERD/DFD), upgrading legacy architectures with **Oracle DB** and **MySQL** to boost data processing efficiency.
- Prototyped UI/UX mockups and supported full-stack enhancements utilizing **Java, PHP**, and **Power BI** dashboards.
- Formulated test plans and led Factory Acceptance Testing (FAT) & User Acceptance Testing (UAT)
- Authored user manuals and functional specifications, bridging technical dev teams and business end-users throughout the **SDLC**.

System Administrator (Intern) | Rifhan Teknologi Sdn. Bhd., Malaysia (2020)

- Maintained high network availability across Ministry of Home Affairs, RMP HQ, and Ministry of Finance with **99% target uptime**. Managing IT desk ticket and enterprise monitoring tools.

Publications & Key Projects

- **Tracing and Correcting Programs (LLM Reasoning):** (*AAAI 2026 Bridge*) & *Master's Thesis*. Enhanced LLM multi-step reasoning using prompt signals and fine-tuning (**PyTorch, QLoRA, FAISS**).
- **Driver Drowsiness Detection System:** *Best Paper Award (ISETech 2023)*. Developed real-time computer vision system via **Transfer Learning & CNNs**. Published in *Journal of Advanced Research in Applied Mechanics*.
- **Voice Synthesis & Sentiment Pipelines:** Built audio matching pipeline (**PyTorch, MelBandRoformer, Gradio, Kafka**) and conducted sentiment analysis on airline customer datasets using **RapidMiner**.

Education, Leadership & Community

Gwangju Institute of Science and Technology (GIST) — M.S. in Artificial Intelligence (Dept. of AI Convergence) | 2024 – 2026

Universiti Teknologi MARA (UiTM) — B.S.(Hons.) in Intelligent Systems Engineering | 2021 – 2023 | GPA: 3.79/4.00 |

Diploma in Computer Science | 2018 – 2021 | GPA: 3.83/4.00

Honors & Awards: Vice Chancellor Award, Dean's Award (Every Semester), Best Paper Award (ISETech 2023).

Spoken Languages: English (IELTS 6.5), Malay, Indonesian, Korean (Conversational - KIIP Level 2).

- **Co-Lead, AI & Data Science Study Group (GIST):** Organized technical seminars, curated ML study materials, and managed venue logistics.
- **Founder & E-Commerce Operator:** Managed end-to-end global cross-border shipping logistics, digital marketing, and customer relationship management.
- **International Student Support Officer (GISA) & Volunteer Tutor:** Guided international student onboarding at GIST.
- **Volunteer Programming Tutor:** Mentored middle and high school students in introductory **Programming Language and Python** and algorithmic problem-solving.